Name: Ahmad Rahdari

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ADDRESS: Rovira i Virgili University, Department of Computer Engineering and Mathematics, Lab 133 DEIM, Avda. Països Catalans 26, 43007 Tarragona, Spain
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Education

Dec. 2025 – Nov. 2028 ***Ph.D. in Computer Engineering and Mathematics***
Universitat Rovira i Virgili, Tarragona, Spain.

- **FPI grant-holder**
- **Working on post-quantum cryptographic protocols and data provenance, e.g., Designated-Commitment TLS (TLS oracles).**
  Supervisor: Dr. Rolando Trujillo Rasua

Sep. 2018 – Feb. 2022 ***M.Sc. in Electrical Engineering - Communication Systems,***
K.N. Toosi University of Technology, Tehran, Iran.

- **Thesis Title: "Proposing a Secure Key Agreement Protocol in Smart Grid Networks based on Elliptic Curve Cryptography"**
  Supervisor: Dr. Bahareh Akhbari
- **Passed Courses:** Advanced Network Security, Information Theory, Channel Coding, Estimation Theory, Advanced Theory of Communications, Numerical Optimization, Stochastic Processes, Digital Communications, Cellular Communications

Sep. 2014 – Sep. 2018 ***B.Sc. in Electrical Engineering***,
K.N. Toosi University of Technology, Tehran, Iran.

- **Thesis Title: "Designing and implementing an intelligent attendance system based on RFID"**
  Supervisor: Dr. Amir Mousavinia
- **Selected Courses:** Communication Systems Principles, Digital Signal Processing, Signals & Systems, Digital Systems, Engineering Mathematics, Numerical Methods, Fundamental of Computer & Programming, Engineering Probability and Statistics, Linear Algebra, Differential Equations

Fields of Interest

- ***Security and Privacy in Computer Systems***
- ***Cryptographic and Decentralized Protocols***
- ***AI for Security and Security for AI***

Publications

Under Review Papers:

• **A. Rahdari**, R. Trujillo-Rasua, A. Belel, E. Nowroozi and S. Gorgin "FogRoam: Cloud-Free Handover Authentication for 6G Vehicular Networks," *IEEE Transactions on Vehicular Technology*.
• **A. Rahdari**, A. Belel and R. Trujillo-Rasua, "Dodona: Post-Quantum TLS Oracle for Data Provenance Attestation," *40th IEEE Computer Security Foundations Symposium (CSF),* 2027.
• E. Nowroozi, **A. Rahdari**, M. Saedi, Y. Habibi, G. Loukas and E. Panaousis, "Confidence-Driven Optimization for Adversarial Transferability in Deep Learning-Based Intrusion Detection Systems," *Neurocomputing*.
• E. Nowroozi, N. Benhammou, **A. Rahdari**, R. Taheri, R. Trujillo-Rasua, M. Waqas, G. Loukas and E. Panaousis, "SecureFedDrone: Watermarked Federated Reinforcement Learning for Secure Multi-Drone Wildlife Monitoring," *IEEE Transactions on Vehicular Technology*.

Published Papers:

• **A. Rahdari** and B. Akhbari, "SKE2SG: A Secure Key Establishment Scheme for Smart Grid Cyber-Physical Systems based on Elliptic Curve Cryptography," *IET Cyber-Physical Systems: Theory & Applications,* 11, 1, 2026. https://doi.org/10.1049/cps2.70045
• E. Nowroozi, M. Mohammadi, **A. Rahdari**, R. Taheri and M. Conti "A Random Deep Feature Selection Approach to Mitigate Transferable Adversarial Attacks," *IEEE Transactions on Network and Service Management,* 2025. https://doi.org/10.1109/TNSM.2025.3594253
• E. Keshavarz, **A. Rahdari** and A. Anand, "A robust dual image watermarking scheme with a decoy watermark for copyright protection and secure authentication," *Multimedia Tools and Applications,* 2025. https://doi.org/10.1007/s11042-025-21020-9
• **A. Rahdari**, E. Keshavarz, E. Nowroozi, R. Taheri, M. Hajizadeh et al., "A Survey on Privacy and Security in Distributed Cloud Computing: Exploring Federated Learning and Beyond," *IEEE Open Journal of the Communications Society,* 2025. https://doi.org/10.1109/OJCOMS.2025.3560034
• **A. Rahdari,** A. Jalili, M. Esnaashari, M. Gheisari, A. A. Vorobeva et al., "Security and Privacy Challenges in SDN-Enabled IoT Systems: Causes, Proposed Solutions, and Future Directions," *Computers, Materials & Continua*, vol. 80, no. 2, pp. 2511-2533. 2024. https://doi.org/10.32604/cmc.2024.052994.
• **A. Rahdari** and MH Tadayon, "Analysis of global trends in providing educational frameworks for training cyber security professionals," *Biannual Journal Monadi for Cyberspace Security (AFTA),* 12(2), pp. 24-32, 2024.

Working Experience

Apr. 2024 – Feb. 2025 ***Remote Research Assistant***

Institute: **University of Greenwich, UK**

Working on "Security for AI and Adversarial Machine Learning" | Supervisor: Dr. Ehsan Nowroozi.

Dec. 2023 – Apr. 2024 ***Remote Research Assistant***

Institute: **Shaoxing University, China**

Working on "privacy preservation and security within IoT networks"

Sep. 2022 - Nov. 2023 ***Researcher***

Institute: **IRAN Telecommunication Research Center (ITRC)**

Working on a project titled "Designing a national roadmap for cyber security specialists development and support" | Supervisor: Dr. Mohammad Hesam Tadayon.

Jun. 2018 - Oct. 2018 ***Electrical Engineering Trainee***

Company: **Raiwan**

Worked on a project titled "Designing an intelligent attendance system based on RFID".

Teaching Experience

Feb. 2021 – Jul. 2022 (3 consecutive semesters) ***Teaching Assistant***, Engineering Probability and Statistics, K.N. Toosi University, Instructor: Dr. Bahareh Akhbari

Feb. 2022 – Jun. 2022 ***Teaching Assistant***, Signals & Systems, K.N. Toosi University, Instructor: Dr. Lotfollah Beygi

Summer 2021 ***Presenting Workshop***, The Art of Technical Presentation, IEEE Iran Section KNTU student branch

Feb. 2021 – Jun. 2021 ***Teaching Assistant***, Differential Equations, K.N. Toosi University, Instructor: Dr. Mojtaba Khalili

Nov. 2020 – Feb. 2021 ***Teaching Assistant***, Engineering Mathematics, K.N. Toosi University, Instructor: Dr. Mojtaba Khalili

Feb. 2020 – Jun. 2020 ***Teaching Assistant***, Fundamentals of Electrical Engineering I, K.N. Toosi University, Instructor: Dr. Parvin Mahmoudabadi

Journal Collaborations & Editorial Contributions

Guest Editor Published SI titled "Security and Privacy in IoT and Smart City: Current Challenges and Future Directions" in the Computers, Materials & Continua Journal (IF: 2).

Reviewer Reviewed 33 manuscripts for 6 journals, including:

- IEEE Transactions on Network and Service Management (14 reviews, IF: 5.4)
- Multimedia Tools and Applications (7 reviews, IF: 3.0)
- Computer Networks (4 reviews, IF: 4.4)
- Vehicular Communications (3 reviews, IF: 5.8)
- International Journal of Communication Systems (3 reviews, IF: 1.8)
- International Journal of Electrical Power and Energy Systems (2 reviews, IF: 5.0)

Selected Academic Projects & Presentations

2024 Clustering-based handwritten digit recognition using Self-Organizing Maps and classification via Single-Layer Feedforward Neural Networks

Using Python | Supervisor: Dr. Mehran Yazdi

Spring 2023 — Secure access control scheme in IoT-enabled Internet of Drones deployment,
Using Avispa and HLPSL | Supervisor: Dr. Ali Hamzeh

Spring 2023 — Conditional Image Synthesis with Auxiliary Classifier GANs,
Using Python | Supervisor: Dr. Eghbal G Mansoori

Fall 2022 — Detection of Sinusoidal Signals with Unknown Parameters in the Presence of White Gaussian Noise,
Using Matlab | Supervisor: Dr. Mostafa Derakhtian

Fall 2019 — Implementation of Berlekamp Algorithm for Decoding Nonbinary RS and BCH Codes,
Using Matlab | Supervisor: Dr. Mahmoud Ahmadian Attari

Spring 2019 — Research and Implementation Challenges in 5G Networks,
Supervisor: Dr. Kamal Mohamed-pour

Spring 2019 — Frequency Estimation and Detection for Sinusoidal Signals with Arbitrary Envelope,
Using Matlab | Supervisor: Dr. Mohammad Ali Sebt

Fall 2018 — Information-Theoretic Key Agreement of Multiple Terminals,
Supervisor: Dr. Bahareh Akhbari

Courses & Certifications

Shiraz University
- ***Neural Network and Deep Learning, Pattern Recognition, Advanced Linear Algebra, Data Security, Detection Theory***, Issued Feb 2024.

IBM
- ***Cyber Threat Intelligence***, Issued Jul 2022, Credential ID: WTFN5HAT4A9L
- ***Penetration Testing, Incident Response and Forensics***, Issued Jul 2022, Credential ID: MJWPS6MRAS8B
- ***IT Fundamentals for Cybersecurity Specialization***
(Including 4 Courses: 1- Introduction to Cybersecurity Tools & Cyber Attacks, 2- Cybersecurity Roles, Processes & Operating System Security 3- Cybersecurity Compliance Framework & System Administration 4- Network Security & Database Vulnerabilities), Issued Jul 2022, Credential ID: 5K67HSS8MFDK

University of Maryland
- ***Hardware Security***, Issued Sep 2021, Credential ID: 5ZJUCVUL3UPV
- ***Usable Security***, Issued May 2021, Credential ID: LNJ9B5YGP6SF

University of Colorado
- ***Applied Cryptography***
(Including 4 Courses: 1- Cryptography and Information Theory 2- Symmetric Cryptography 3- Asymmetric Cryptography and Key Management 4- Cryptographic Hash and Integrity Protection), Issued May 2021, Credential ID: K98TQNGE4URS
- ***Mathematical Foundations for Cryptography***, Issued May 2021, Credential ID: TGFXJCZBQPPR

Vanderbilt University
- ***Introduction to Programming with MATLAB***, Issued Sep 2021, Credential ID: SGU7ZJNK7Z5W

Honors & Awards

2025-2028	FPI predoctoral grant-holder (Spain)
2021	Winner of the 9th National Debate of Iranian Students, first among about 5000 students (National Gold Medal).
2018 - 2021	Own Tuition Wavier for studying the M.Sc. at K.N. Toosi University of Technology.
2018	Ranked 118st among more than 20,000 participants of the nationwide post-graduate entrance exam in the field of Electrical Engineering.
2014 - 2018	Own Tuition Wavier for studying the B.Sc. at K.N. Toosi University of Technology.
2014	Top %0.5 among approximately 300,000 participants of the nationwide university entrance exam in the field of Mathematics and Physics.
2013	Selected from the first stage of the 31th Iranian Mathematical Olympiad, Top %1 among over 90,000 participants.

Computer Skills

Languages	Expert in MATLAB and High-Level Protocol Specification Language (HLPSL). Familiar with Python, and C.
Softwares	Expert in AVISPA tool. Familiar with Tamarin Prover, Visual Studio, and Scyther tool.
Version Control	Familiar with Git.
Operating System	Using Windows, Familiar with Ubuntu.
Document preparation	Expert in Microsoft Office, Overleaf, and LaTex.

Language Skills

English: Fluent **Persian**: Native

References

- **Dr. Rolando Trujillo Rasua**: rolando.trujillo@urv.cat
  Rovira i Virgili University, Spain
  Ramon i Cajal Researcher

- **Dr. Ehsan Nowroozi**: e.nowroozi@greenwich.ac.uk
  University of Greenwich, UK
  Senior Lecturer (Associate Professor)